

A Packing of 9 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 9 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.999732. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.962710, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 9 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.999732\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 9$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.962710, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 9/8 = 3.000000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N9.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
149986620243152302343239578864097032974679969401755708599014521013345502
351038599191770326593484561762263283690930566020132636495102932307792323
2617949151653279125866760269122567889681059720383
```

denominator:

```
499999999998599291785704362331812300190691913910579502386657632351251311
100283952450896075792796798703218696537772598547128251034054898380432378
420350242481818858428695512006290890581407754035
```

B Coordinates

9 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N9.txt`.

```
438173219560 307187227102 361424826440 8450075254 -91110123015 98869805254 433124755104 -2573457355467983079865 321157472736 407851431413 360123754499
125889588648 61537882594 -546836082790 38927416217611409367994 -343087398100 -103878684762 1291138544390379034484 -355403171967 -67888922169 77079895700
460812719078 6019233691 -238229653643 -498997540645-123114599801 296843430973 -239581201063 7138540571766204455367 -302861261567 411758644756 193614337351
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-457424525146 -89753816439 -439615700874 -337200783465313207233 369292974812 -33393714002 -1846391012583932862203 -260063434659 59228348308 170124939482
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```